

Chapter 2 - Basic Laws

- L#1 { 2.2 Ohm's Law.
2.3 Nodes, Branches, and Loops.
- L#2 { 2.4 Kirchhoff's Laws.
- L#3 { 2.5 Series Resistors and Voltage Division.
2.6 Parallel Resistors and Current Division.
- L#4 { 2.7 Wye-Delta Transformations.



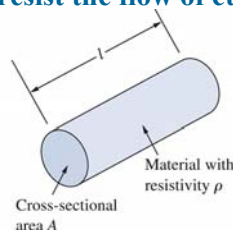
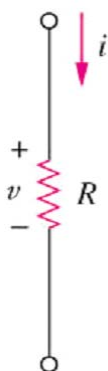
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Lecture #1

2.2 Ohms Law: $V = IR$

- Ohm's law states that the voltage V across a resistor is directly proportional to the current I flowing through the resistor. $v \propto i$
 $v = Ri$
- Resistance R : Physical property of a material to resist the flow of current.



$$R = \rho \frac{l}{A} = \frac{V}{I}$$



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2.2 Ohms Law

Resistivity ρ & Resistance R

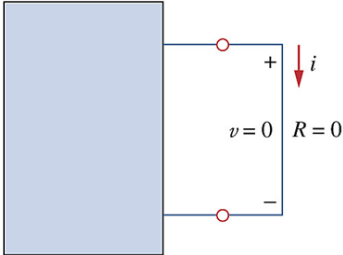
Material	Resistivity ($\Omega\cdot\text{m}$)	Usage
Copper	1.72×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semi-conductor
Silicon	6.4×10^2	Semi-conductor
Paper	1×10^{10}	Insulator
Teflon	3×10^{12}	Insulator

$$R = \rho \frac{l}{A} = \left[\Omega \cdot \text{m} \frac{\text{m}}{\text{m}^2} \right] = [\Omega]$$

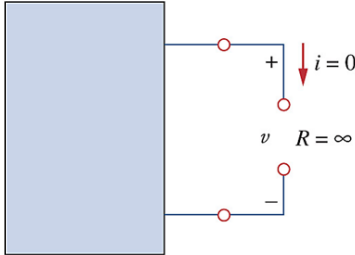
Resistors

2.2 Ohms Law

Resistance R - extremes



$R = 0$: SHORT Circuit (SC)
 $v = iR = i \times 0 = 0$



$R = \infty$: OPEN Circuit (OC)
 $i = \lim_{R \rightarrow \infty} \frac{v}{R} = 0$

$$R = \frac{V}{I}$$



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Ohms Law

Conductance G

- Conductance is the ability of an element to conduct electric current; it is the reciprocal of resistance R and is measured in mho's [U] or Siemens [S].

$$G = \frac{1}{R} = \frac{i}{v} [S]$$

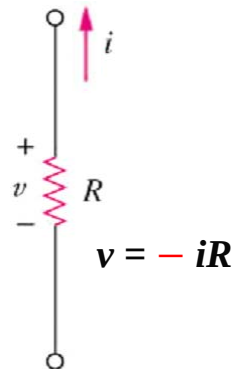
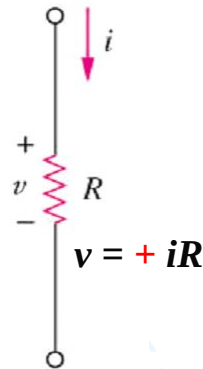
$$i = Gv$$



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Passive Sign Convention



Power dissipated in a Resistor

Diagram showing a resistor R with current i flowing downwards and voltage v positive at the top.

$$p = +iv \begin{cases} \xrightarrow{v=+iR} = +i(+iR) = +i^2R \\ \xrightarrow{i=v/R} = +\frac{v}{R}v = +\frac{v^2}{R} \end{cases}$$

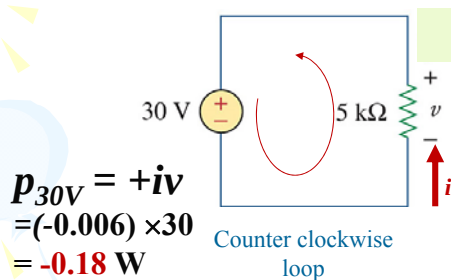
MUCHO IMPORTANTE:
Since i^2 , v^2 & R is always ≥ 0 :
 P in a resistor is **ALWAYS ≥ 0** (positive)

Diagram showing a resistor R with current i flowing upwards and voltage v positive at the top.

$$p = -iv = -i(-iR) = i^2R$$

Example 2-2

In the circuit shown, calculate the current i , the conductance G and the power p .



$$p_{30V} = +iv$$

$$= (-0.006) \times 30$$

$$= -0.18\text{ W}$$

delivered

$$i = -v/R$$

Negative bc of passive sign convention

$$= -30 / 5 \times 10^3$$

$$= -6\text{ mA}$$

$$p_R = -iv$$

$$= -(-0.006) \times 30$$

$$= 0.18\text{ W (absorbed)}$$

or:

$$p_R = i^2 R$$

$$= (6 \times 10^{-3})^2 \times 5 \times 10^3$$

$$= 180\text{ mW}$$

or:

$$p_R = v^2 / R$$

$$= (30)^2 / 5000$$

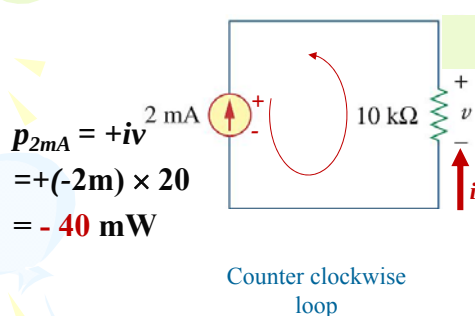
$$= 180\text{ mW}$$



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PP 2-2: For the circuit shown, calculate the voltage v , conductance G and power p .



$$p_{2mA} = +iv$$

$$= +(-2\text{ m}) \times 20$$

$$= -40\text{ mW}$$

$$i = -2\text{ mA}$$

Negative bc i is in opposite direction of 2 mA source current

$$v = -iR$$

$$= -(-2 \times 10^{-3}) \times 10 \times 10^3$$

$$= 20\text{ V}$$

$$p_R = -iv$$

$$= -(-2\text{ m}) \times 20$$

$$= 40\text{ mW}$$

or:

$$p_R = i^2 R$$

$$= (-2 \times 10^{-3})^2 \times 10 \times 10^3$$

$$= 40\text{ mW}$$

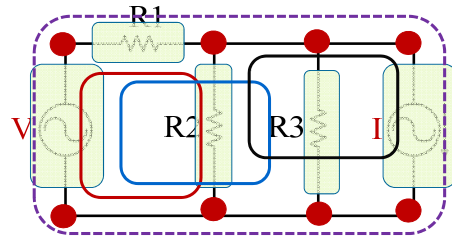


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2.3 Nodes, Branches and Loops



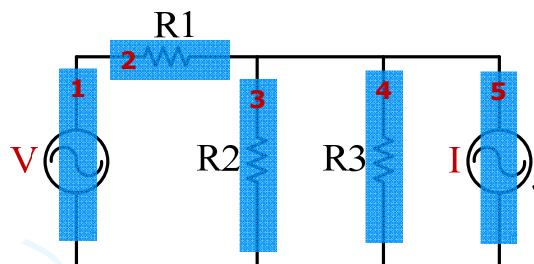
- A **branch** represents a single element such as a voltage source or a resistor.
- A **node** is the point of connection between two or more branches.
- A **loop** is any closed path in a circuit.

BRANCHES

"A single element such as a voltage source or a resistor."

How many branches?

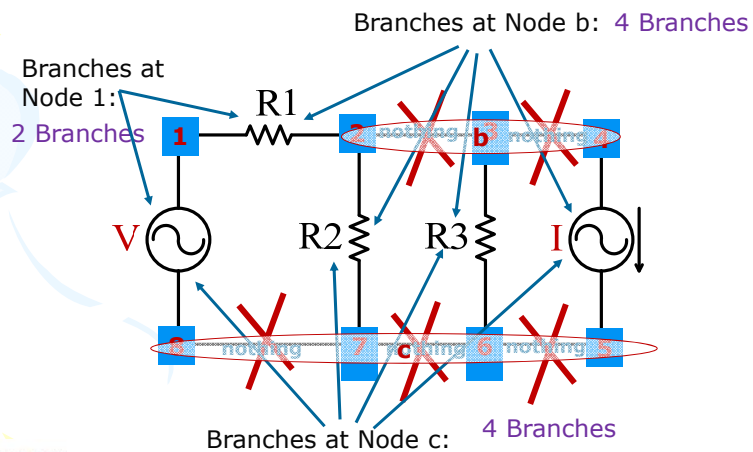
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Nodes

"The point of connection between two or more branches."

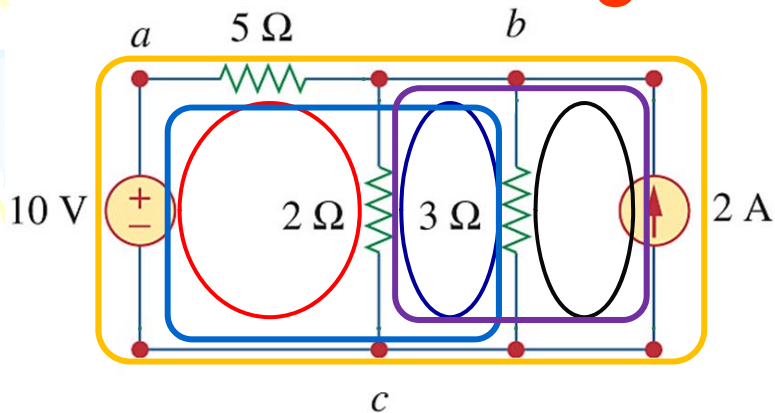
How many nodes? ~~wrong 8~~ **3**



LOOPS

"Any closed path in a circuit."

How many loops? **6**

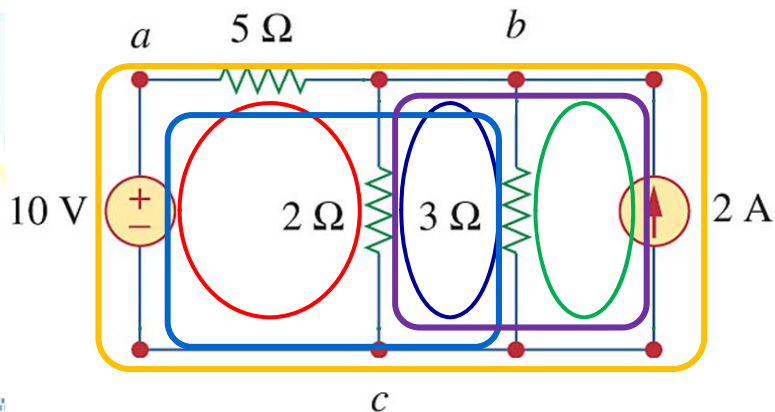


INDEPENDENT LOOPS

"A loop that contains at least one branch which is not a part of any other independent loop."

"A loop that cannot be formed by combining / merging any of the other loops."

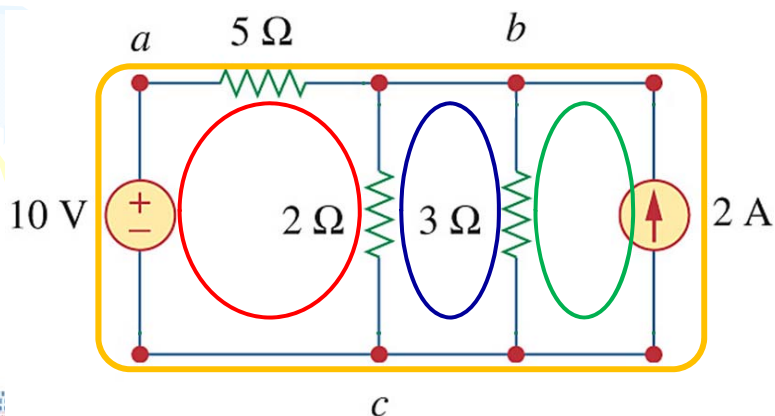
Independent sets leads to independent equations



When is it NOT an independent set?

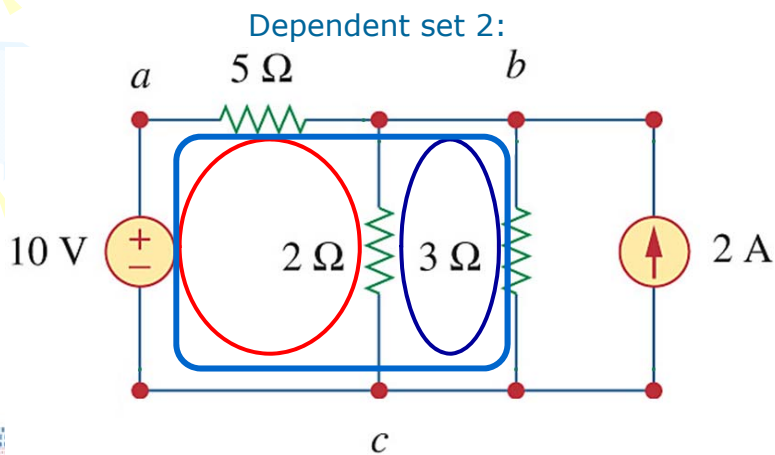
If any loop can be formed by merging any of the other loops.

Dependent set 1:



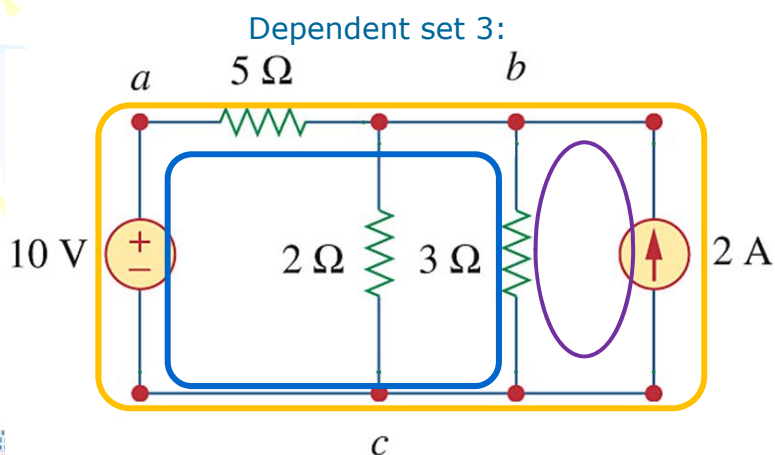
When is it NOT independent?

If any loop can be formed by merging any of the other loops.



When is it NOT independent?

If any loop can be formed by merging any of the other loops.



PP 2-4: How many branches, nodes and independent loops does the circuit have?

Branches = **5**

Nodes = **3**

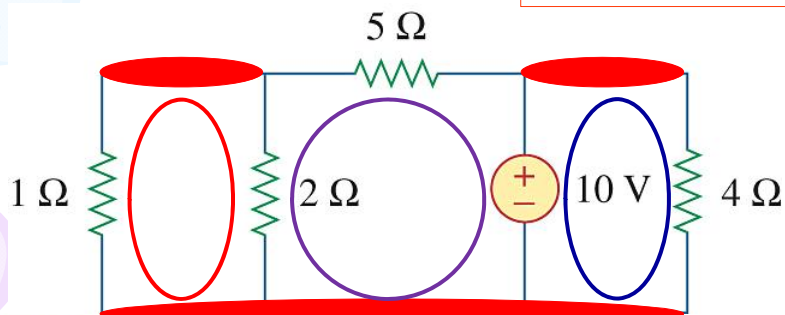
Finding loops mathematically?

Loops = **3**

$$b = l + n - 1$$

$$5 = l + 3 - 1$$

$$l = 3$$



Problem 2.5: Find the number of nodes, branches and independent loops.

Branches = **15**

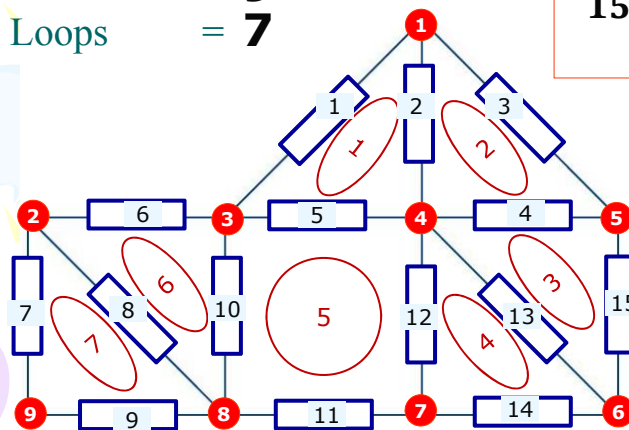
Nodes = **9**

Loops = **7**

$$b = l + n - 1$$

$$15 = l + 9 - 1$$

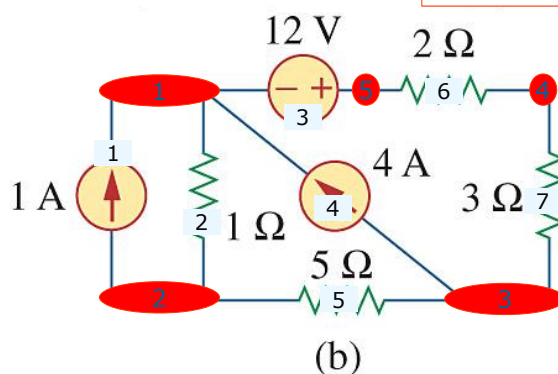
$$l = 7$$



Problem 2.7b: Find the number of nodes, branches and independent loops.

Branches = **7**
 Nodes = **5** ~~wrong~~
 Loops = **3**

$$\begin{aligned} b &= l + n - 1 \\ 7 &= l + 5 - 1 \\ l &= 3 \end{aligned}$$



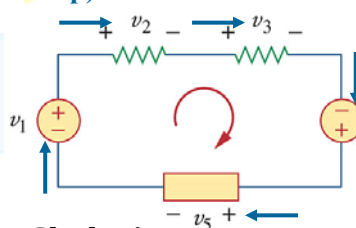
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Lecture #2: 2.4 Kirchhoff's Laws

Kirchhoff's Voltage Law (KVL)

- Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

$$\sum_{m=1}^M v_m = 0$$

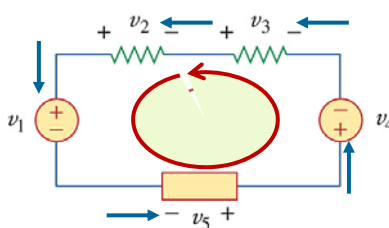


Clock wise

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

Convention:

- Always go clock wise
- In at +: ADD
- In at -: SUBTRACT



Counter clock wise

$$\begin{aligned} +v_1 - v_5 + v_4 - v_3 - v_2 &= 0 \\ -v_1 + v_2 + v_3 - v_4 + v_5 &= 0 \end{aligned} \quad \} \times -1$$

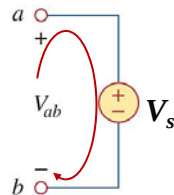
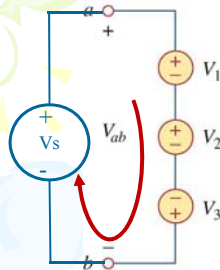


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Kirchhoff's Voltage Law (KVL)

Voltage sources in series

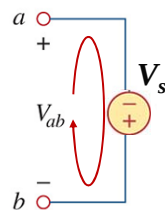


$$-V_{ab} + V_s = 0$$

$$V_{ab} = V_s$$

$$-V_s + V_1 + V_2 - V_3 = 0$$

$$V_s = V_1 + V_2 - V_3$$



$$-V_{ab} - V_s = 0$$

$$V_{ab} = -V_s$$



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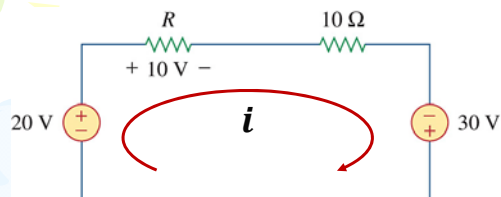
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P 2-37 Find R for the circuit.

Ohm's Law:

$$+iR = 10$$



$$\text{KVL: } -20 + 10 + 10i - 30 = 0$$

$$i = \frac{40}{10} = 4 \text{ A}$$

$$+4R = 10$$

$$R = \frac{10}{4} = 2.5 \Omega$$

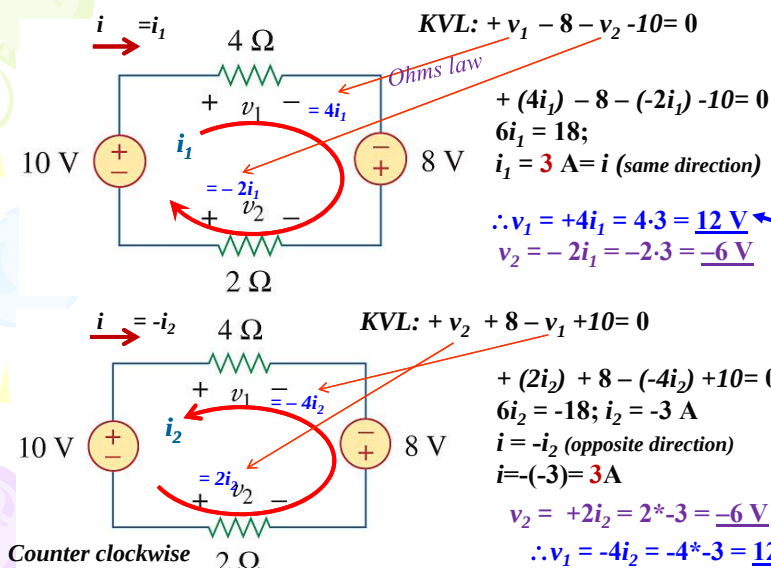


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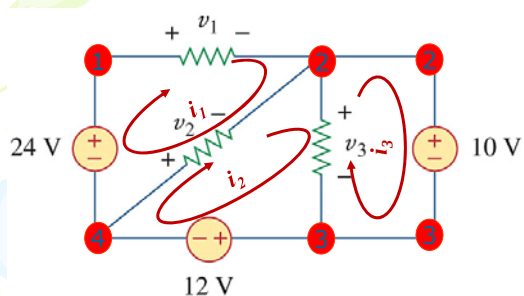
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PP 2-5: Find i , v_1 and v_2



P 2-17: Determine v_1 , v_2 and v_3 in the circuit.



Nodes=4
Branch=6
Loops required: $L=B+1-n$
 $L=6+1-4=3$

KVL@ i_1 :

$v_1 - v_2 - 24 = 0$ (1)

KVL@ i_2 :

$v_3 + 12 + v_2 = 0$ (2)

KVL@ i_3 :

$10 - v_3 = 0$

$\Rightarrow v_3 = 10 \text{ V}$

Substitute v_3 in (2)

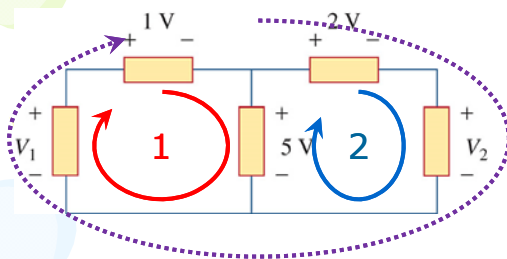
$10 + 12 + v_2 = 0$;

$v_2 = -22 \text{ V}$

Substitute v_2 in (1)

$v_1 - (-22) - 24 = 0$;

$v_1 = 2 \text{ V}$

Problem 2-11: Calculate V_1 and V_2 .**KVL2:**

$$+ V_2 - 5 + 2 = 0$$

$$\therefore V_2 = 3 \text{ V}$$

KVL1:

$$- V_1 + 1 + 5 = 0$$

$$\therefore V_1 = 6 \text{ V}$$

Nodes=4

Branch=5

Loops required: $l=B+1-n$ $L=5+1-4=2$ **The Test:...**

$$- V_1 + 1 + 2 + V_2 = 0$$

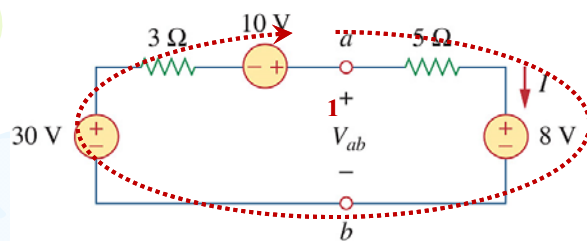
$$- 6 + 1 + 2 + 3 = 0$$

$$- 6 + 6 = 0$$

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Problem 2-18: Determine I and V_{ab} .**I:** same direction as that of loop current**KVL1:**

$$- 30 + v_3 - 10 + v_5 + 8 = 0$$

$$- 30 + 3I - 10 + 5I + 8 = 0$$

$$\therefore 8I = 32$$

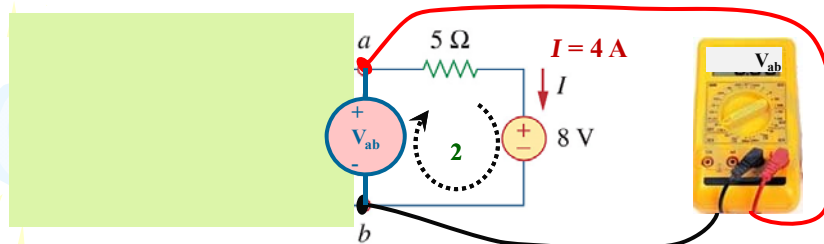
$$I = 4 \text{ A}$$

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Method#1: Determine V_{ab}



KVL2:

$$\begin{aligned} -V_{ab} + 5I + 8 &= 0 \\ \therefore V_{ab} &= 5(4) + 8 \\ &= 28 \text{ V} \end{aligned}$$

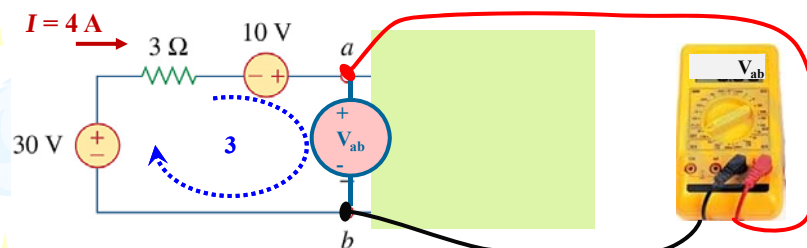


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Method#2 Determine V_{ab}



KVL3:

$$\begin{aligned} -30 + 3I - 10 + V_{ab} &= 0 \\ \therefore V_{ab} &= 40 - 3(4) \\ &= 28 \text{ V} \end{aligned}$$



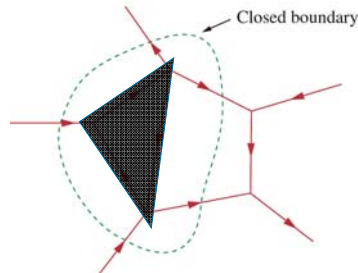
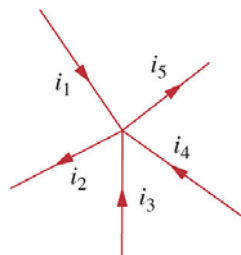
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KCL-Kirchhoff's Current Law

- Kirchhoff's **current law (KCL)** states that the algebraic sum of currents entering and leaving a node (or a closed boundary) is zero.



Mathematically
$$\sum_{n=1}^N i_n = 0$$

N = Number of branches connected to the node.

i_n = n^{th} current entering or leaving.



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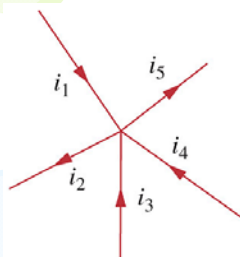
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Kirchhoff's Current Law (KCL)

"What goes in, must come out"

Current entering a node: ADD +

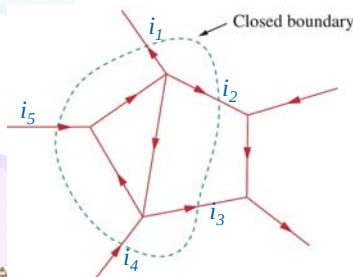
Current leaving a node: SUBTRACT -



$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$$

$$I_{\text{Entering}} = I_{\text{Leaving}}$$

$$i_1 + i_3 + i_4 = i_2 + i_5$$



Closed Boundary (Super Node):

$$-i_1 - i_2 - i_3 + i_4 + i_5 = 0$$



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Kirchhoff's Current Law (KCL) Current Sources in Parallel

$I_T = I_{S1}$ (same direction)
 $I_{S1} = +I_1 - I_2 + I_3$

$I_T = -I_{S2}$ (opposite direction)
 $I_{S2} = -I_1 + I_2 - I_3$

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Kirchhoff's Current Law (KCL) Resistive networks – different current direction

KCL @ node (a) $\sum i = 0$
 $-i_1 - i_2 + i_3 = 0$
 Or $I_{in} = I_{out}$
 $i_3 = i_1 + i_2$

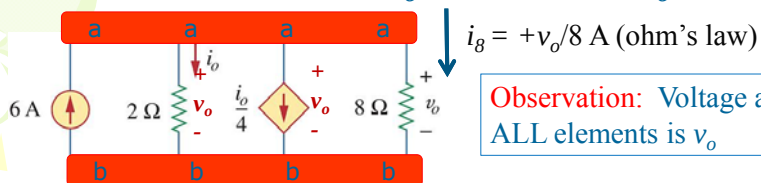
KCL @ node (b)
 $+i_1 - i_2 + i_3 = 0$
 Or $I_{in} = I_{out}$
 $i_1 + i_3 = i_2$

KCL @ node (c)
 $+i_1 + i_2 + i_3 = 0$
 Or $I_{in} = 0$
 $i_1 + i_2 + i_3 = 0$

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PP 2-7: Find current i_o and voltage v_o .



$$i_g = +v_o/8 \text{ A (ohm's law)}$$

Observation: Voltage across ALL elements is v_o

Ohm's law:

$$i_o = +v_o/2$$

$$\therefore v_o = 2i_o$$

The Test:...

$$+6 - i_o - i_o/4 - v_o/8 = 0$$

$$+6 - 4 - 4/4 - 8/8 = 0$$

$$6 - 4 - 1 - 1 = 0$$

KCL@ node (a): $\sum i = 0$:

$$+6 - i_o - i_o/4 - i_g = 0$$

$$+6 - i_o - i_o/4 - v_o/8 = 0$$

$$+6 - i_o - i_o/4 - (2i_o)/8 = 0$$

$$i_o = \underline{4 \text{ A}}$$

$$v_o = 2 * 4 = \underline{8 \text{ V}}$$

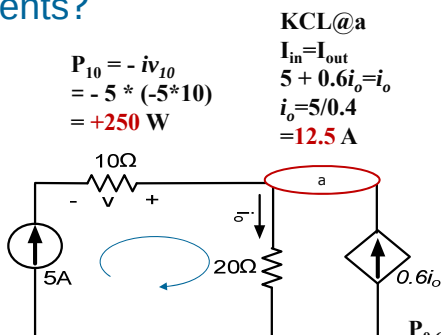


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Calculate the current i_o , power associated with all the elements?



$$P_{10} = -i v_{10}$$

$$= -5 * (-5 * 10)$$

$$= +250 \text{ W}$$

KCL@a

$$I_{in} = I_{out}$$

$$5 + 0.6i_o = i_o$$

$$i_o = 5/0.4$$

$$= 12.5 \text{ A}$$

$$P_5 = -5 * v_5$$

KVL

$$-v_5 - v + v_{20} = 0$$

$$v_5 = -(-5 * 10) + (12.5 * 20)$$

$$= (50) + (250) = 300 \text{ V}$$

$$P_5 = -5 * 300$$

$$= -1500 \text{ W}$$

$$P_{20} = +i_o v_{20}$$

$$= 12.5 * (12.5 * 20)$$

$$= +3125 \text{ W}$$

$$P_{0.6i_o} = -i_o v_{0.6i_o}$$

$$= -0.6i_o * v_{20}$$

$$= -0.6(12.5) * (12.5 * 20)$$

$$= -1875 \text{ W}$$

Verify: $\sum P = 0$

$$P_{10} + P_5 + P_{20} + P_{0.6i_o} = 0$$

$$+250 - 1500 + 3125 - 1875$$

$$= 0$$

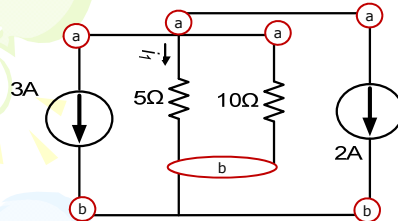


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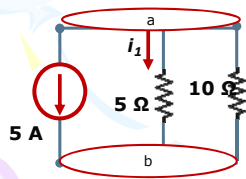
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Calculate the current i_1 ?



Since there are only two nodes "ab"
All elements are parallel
3A & 2A are in same direction
 $I_s = 3 + 2 = 5 \text{ A}$



i_1 using current divider

$$i_1 = -5 * \left(\frac{\frac{1}{5}}{\frac{1}{5} + \frac{1}{10}} \right) = -3.33 \text{ A}$$

-ve bc 5A & i_1 are in opposite direction



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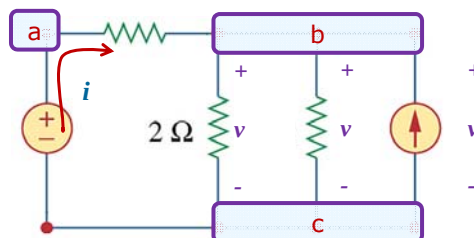
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Lecture #3 Series and Parallel Resistors

SERIES

Two (2) elements are in series if:

1. They exclusively share a single node (node **a**)
2. $\therefore$ carry the **same current**. (i)



PARALLEL

Two or more elements are in parallel if:

1. They are connected to the same two nodes
2. $\therefore$ have the **same voltage**, (v)



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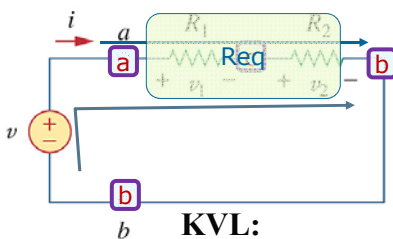
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2.5 Series Resistors

- Series:** Two or more elements are in series if they are cascaded or connected sequentially and consequently carry the same current.

Purpose is to replace all series resistance with single resistor



R_1 & R_2 are in Series

$$v_1 = +iR_1 \text{ or } i = +\frac{v_1}{R_1}$$

$$v_2 = +iR_2 \text{ or } i = +\frac{v_2}{R_2}$$

KVL:

$$-v + v_1 + v_2 = 0$$

$$v = v_1 + v_2$$

$$= i(R_1 + R_2)$$

$$v = iR_{eq}$$

$$i = \frac{v}{R_1 + R_2}$$

$$i = \frac{v}{R_{eq}}$$



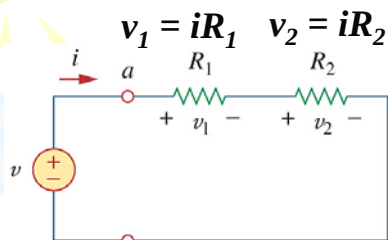
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VOLTAGE DIVISION resistors in series

Find v_1 & v_2 given v and all R 's

Calculating individual resistor's voltage without calculating the current



$$v_1 = iR_1 \quad v_2 = iR_2$$

$$i = \frac{v}{R_1 + R_2}$$

$$i = \frac{v}{R_{eq}}$$

Substitute i into v_1 $v_1 = \frac{v}{R_1 + R_2} R_1 = \frac{R_1}{R_{eq}} v$

Substitute i into v_2 : $v_2 = \frac{v}{R_1 + R_2} R_2 = \frac{R_2}{R_{eq}} v$



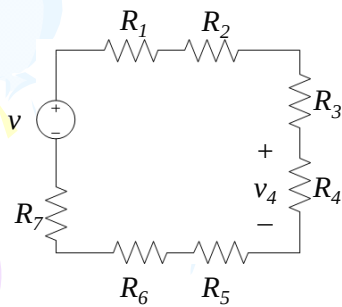
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2.5 Series Resistors & Voltage Division

R_{eq} for more than 2 R's

$$R_{eq} = R_1 + R_2 + R_3 + R_4 + R_5 + R_6 + R_7 = \sum_{n=1}^7 R_n$$



$$v_4 = \frac{??}{R_{eq}} v$$

$$v_6 = \frac{??}{R_{eq}} v$$



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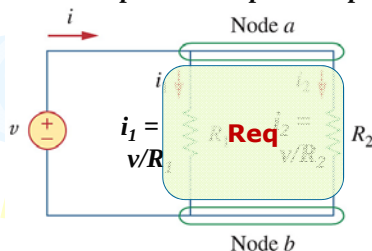
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2.6 Parallel Resistors & Current Division

- Parallel:** Two or more elements are in parallel if they are connected to the same two nodes and consequently have the same voltage across them.

Purpose is to replace all parallel resistance with single resistor



$$\text{KCL@}a: i - i_1 - i_2 = 0$$

$$i = i_1 + i_2 \quad (1)$$

Substitute i_1 and i_2 into (1):

$$i = \frac{v}{R_1} + \frac{v}{R_2} = v \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = \frac{v}{R_{eq}}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

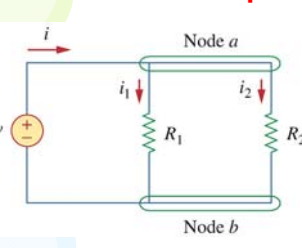


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R_{eq} for Parallel Resistors

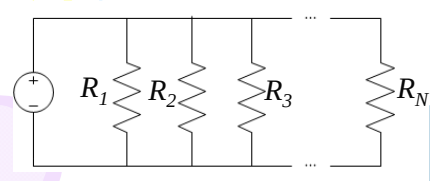


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_1 + R_2}{R_1 R_2}$$

For **TWO** // resistors:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = R_{||}$$

Then does it mean that:



$$R_{eq} = \frac{R_1 R_2 R_3 \dots R_N}{R_1 + R_2 + R_3 + \dots + R_N}$$

NO!

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N}$$


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2.6 Parallel Resistors & Current Division

Find i_1 & i_2 . Given i and R

Calculating individual resistor's current without calculating the voltage

Two resistors in parallel

$$v = i R_{eq} = i \frac{R_1 R_2}{R_1 + R_2}$$

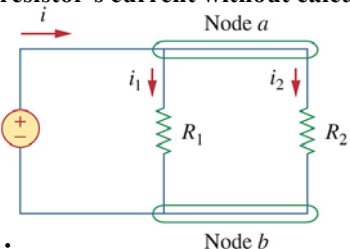
Ohms Law

$$v = i_1 R_1$$

Substituting v for i_1 :

$$i_1 \cancel{R_1} = i \frac{\cancel{R_1} R_2}{R_1 + R_2}$$

$$i_1 = i \frac{R_2}{R_1 + R_2}$$



Ohms Law

$$v = i_2 R_2$$

Substituting v for i_2 :

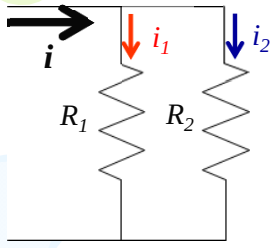
$$i_2 \cancel{R_2} = i \frac{R_1 \cancel{R_2}}{R_1 + R_2}$$

$$i_2 = i \frac{R_1}{R_1 + R_2}$$

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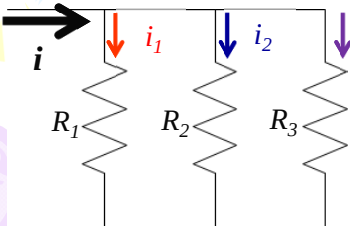
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Current Division



$$i_1 = i \frac{R_2}{R_1 + R_2}$$

$$i_2 = i \frac{R_1}{R_1 + R_2}$$



$$i_2 = ?$$

$$i_2 = i \frac{R_1 R_3}{R_1 + R_2 + R_3}$$

✗

$$i_2 = i \frac{G_2}{G_1 + G_2 + G_3}$$

✓

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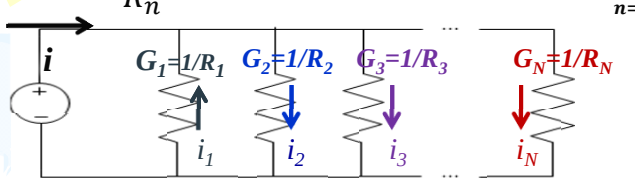
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Current Division – more than 2 R's in ||

Use Conductance (G) instead of resistance (R)

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_N}$$

$$G_n = \frac{1}{R_n} \quad \therefore G_{eq} = G_1 + G_2 + G_3 + \dots + G_N = \sum_{n=1}^N G_n$$



$$i_1 = -i \frac{G_1}{G_{eq}}$$

$$i_2 = i \frac{G_2}{G_{eq}}$$

$$i_3 = i \frac{G_3}{G_{eq}}$$

$$i_N = i \frac{G_N}{G_{eq}}$$

Currents i & i_1 are in opposite direction

$$i_n = \frac{G_n}{G_1 + G_2 + G_3 + \dots + G_N} i$$

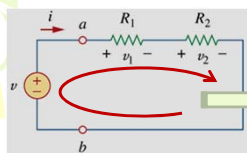
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Summary

Series Resistors



$$v_1 = iR_1 \text{ and } v_2 = iR_2$$

$$R_{eq} = R_1 + R_2$$

$$R_{eq} = R_1 + R_2 + \dots + R_n$$

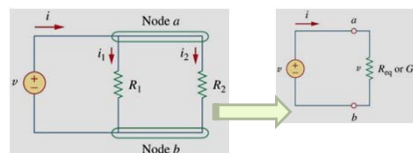
Voltage divider:

$$v_1 = v \cdot \frac{R_1}{R_1 + R_2} \text{ and}$$

$$v_2 = v \cdot \frac{R_2}{R_1 + R_2}$$

$$v_n = v \cdot \frac{R_n}{R_1 + R_2 + \dots + R_n}$$

Parallel Resistors



$$i_1 = \frac{v}{R_1}; i_2 = \frac{v}{R_2} \quad \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

$$G_{eq} = G_1 + G_2 + \dots + G_N$$

Current divider: $i_1 = i \cdot \frac{G_1}{G_{eq}}$ and

$$i_2 = i \cdot \frac{G_2}{G_{eq}}$$

$$i_n = i \cdot \frac{G_n}{G_1 + G_2 + \dots + G_N}$$

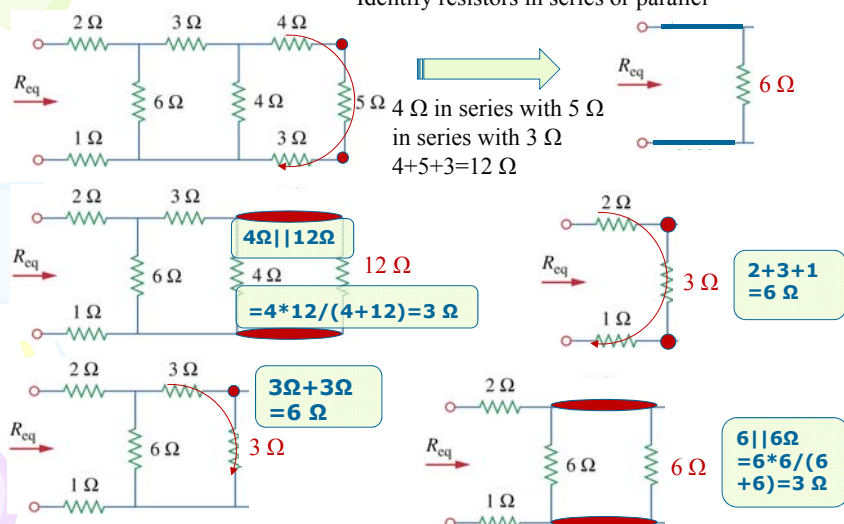


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PP 2-9 Determine R_{eq}

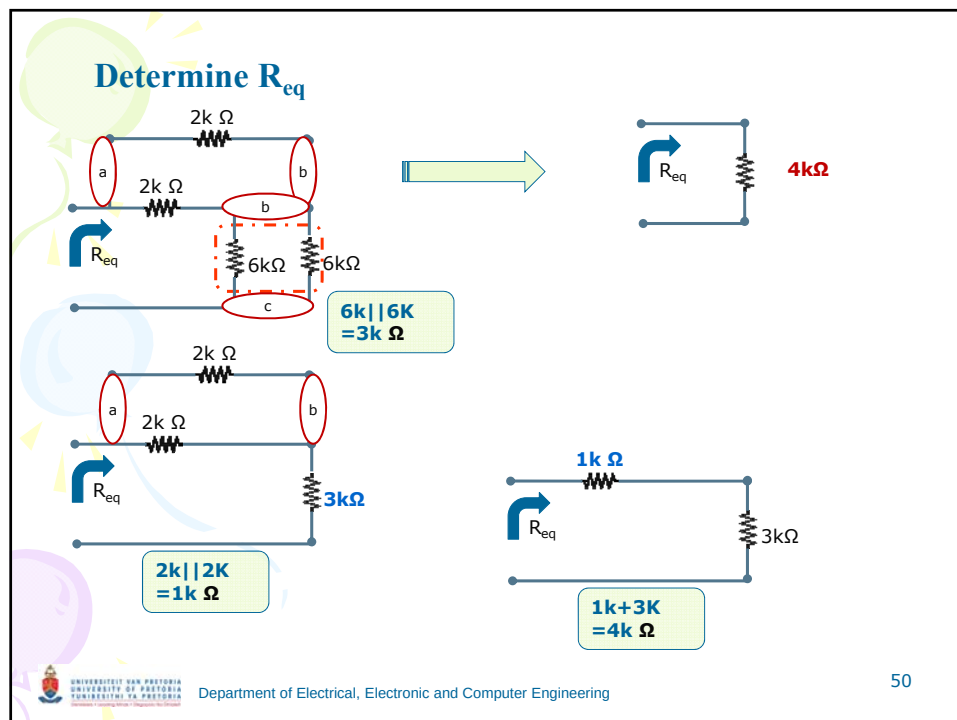
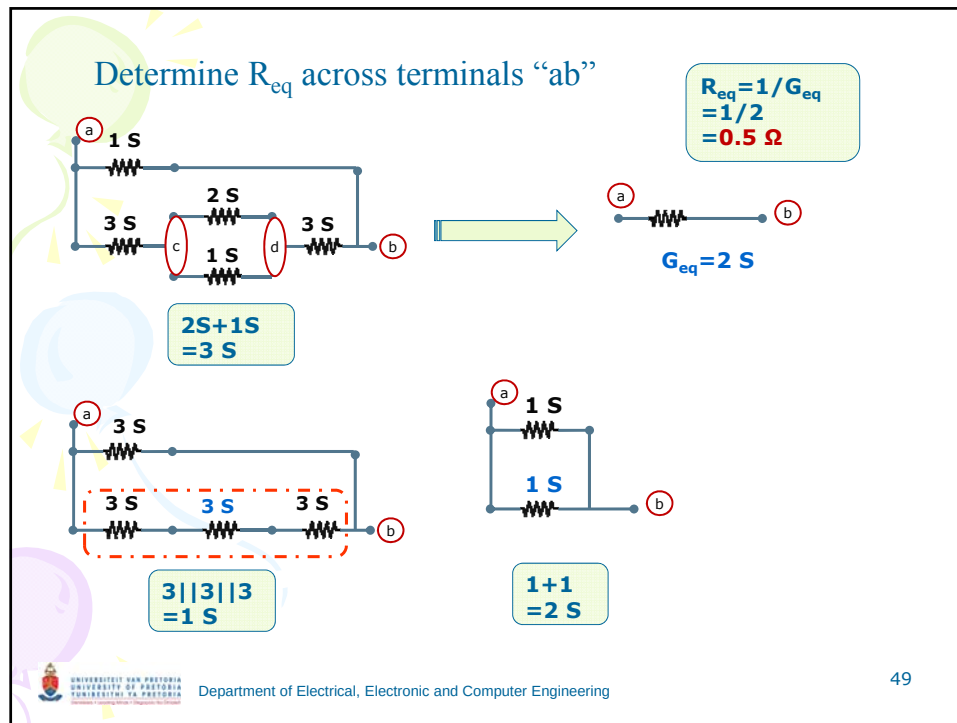
Identify resistors in series or parallel



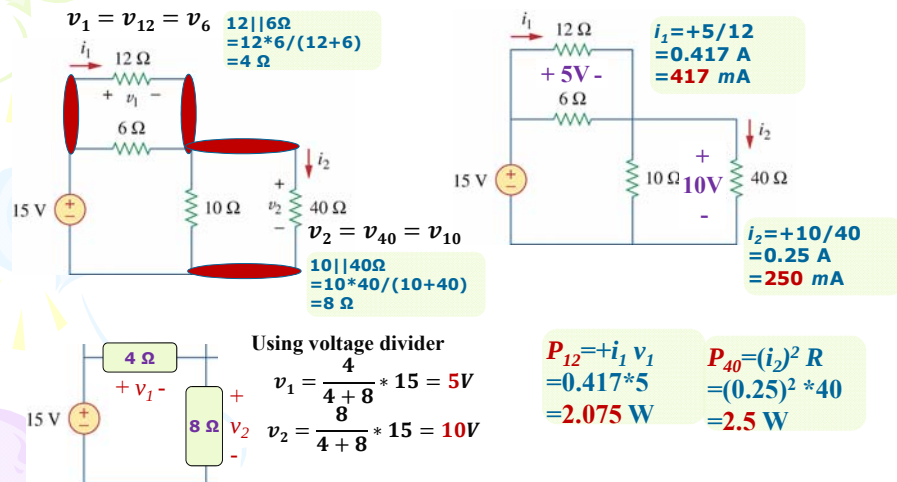
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PP 2-12 Find v_1 and v_2 in the circuit. Also calculate i_1 and i_2 and the power dissipated in the $12\ \Omega$ and $40\ \Omega$ resistors.



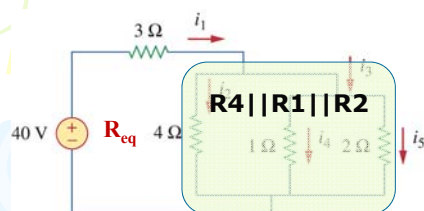
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P 2-31 Determine i_1 to i_5 .

Method #1: using resistors $i_1 = \frac{40}{R_{eq}}$



$$R_{eq} = 3 + R_{4||1||2}$$

$$\frac{1}{R_{4||1||2}} = \frac{1}{4} + \frac{1}{1} + \frac{1}{2}$$

$$R_{4||1||2} = 1 / \left(\frac{1}{4} + \frac{1}{1} + \frac{1}{2} \right) = 0.57\ \Omega$$

$$R_{eq} = 3 + 0.57 = 3.57\ \Omega$$

$$i_1 = \frac{v}{R_{eq}} = \frac{40}{3.57} = 11.2\ \text{A}$$

use current divider to calculate i_2, i_3

$$i_2 = \frac{R_{1||2}}{4 + R_{1||2}} \cdot 11.2$$

$$R_{1||2} = \frac{1 \cdot 2}{1 + 2} = 0.67\ \Omega$$

$$i_2 = \frac{0.67}{4.67} \cdot 11.2 = 1.6\ \text{A}$$

$$i_3 = i_1 - i_2 = 11.2 - 1.6 = 9.6\ \text{A}$$

$$i_4 = \frac{2}{1 + 2} \cdot i_3$$

$$i_4 = \frac{2}{1 + 2} \cdot 9.6 = 6.4\ \text{A}$$

$$i_5 = 9.6 - 6.4 = 3.2\ \text{A}$$

Advice: use current divider with Conductor, G



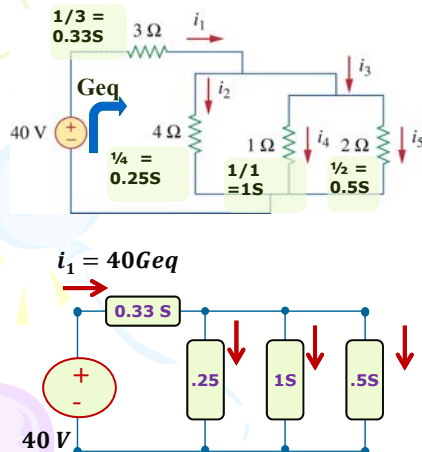
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P 2-31 Determine i_1 to i_5 .

Method #2: Using Conductance G



1. Calculate the conductance

$$G_{eq} = 0.33 \parallel (0.25 + 1 + 0.5) \\ = 0.33 \parallel 1.75 \\ = 0.28 \text{ S}$$

$$i_1 = v * G_{eq} = 40 * 0.28 \\ = 11.2 \text{ A}$$

2. use current divider to find i 's

$$i_n = i_1 \cdot \frac{G_n}{G_1 + G_2 + \dots + G_N}$$

$$i_2 = \frac{0.25}{1.75} * 11.2 = 1.6 \text{ A}$$

$$i_3 = i_1 - i_2 = 11.2 - 1.6 = 9.6 \text{ A}$$

$$i_4 = \frac{1}{1.75} * 11.2 = 6.4 \text{ A}$$

$$i_5 = \frac{1/2}{1.75} * 11.2 = 3.2 \text{ A}$$



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P 2-32 Determine i_1 , i_2 , i_3 and i_4 in the circuit.

Current divider when two resistor in parallel:

$$10 \parallel 40 = R_a = 8 \Omega$$

$$20 \parallel 30 = R_b = 12 \Omega$$

$$i_a = i \cdot \frac{R_b}{R_a + R_b} \\ = 20 * \frac{12}{8 + 12} \\ = 12 \text{ A}$$

$$i_b = i \cdot \frac{R_a}{R_a + R_b} \\ = 20 * \frac{8}{8 + 12} \\ = 8 \text{ A}$$

$$i_3 = i_a \cdot \frac{10}{10 + 40} = 12 * 0.2 \\ = 2.4 \text{ A}$$

$$i_4 = i_a \cdot \frac{40}{10 + 40} = 12 * 0.8 \\ = 9.6 \text{ A}$$

$$i_1 = i_b \cdot \frac{20}{20 + 30} = 8 * \frac{20}{50} \\ = 3.2 \text{ A}$$

$$i_2 = i_b \cdot \frac{30}{20 + 30} = 8 * 0.6 \\ = 4.8 \text{ A}$$



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P 2-32 Determine i_1, i_2, i_3 and i_4 in the circuit. Method#2

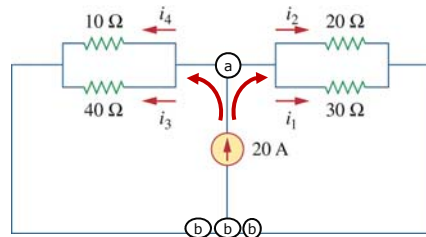
Current divider using Conductance G:

$$G_{eq} = G_1 + G_2 + \dots + G_N$$

$$i_n = i \cdot \frac{G_n}{G_1 + G_2 + \dots + G_N}$$

$$G_{eq} = G_1 + G_2 + G_3 + G_4 = \frac{1}{30} + \frac{1}{20} + \frac{1}{40} + \frac{1}{10} = \frac{25}{120}$$

All the resistance are in parallel as they have common node "ab"



$$i_1 = +20 \cdot \frac{1/30}{25/120} = 3.2 \text{ A}$$

$$i_2 = +20 \cdot \frac{1/20}{25/120} = 4.8 \text{ A}$$

$$i_3 = +20 \cdot \frac{1/40}{25/120} = 2.4 \text{ A}$$

$$i_4 = +20 \cdot \frac{1/10}{25/120} = 9.6 \text{ A}$$



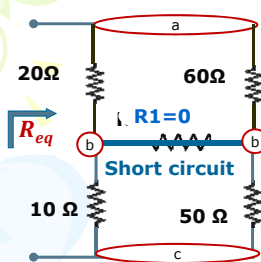
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Calculate R_{eq} , when $R_1 = 0 \Omega$; $R_1 = \infty$

Identify if any resistance in series or parallel?

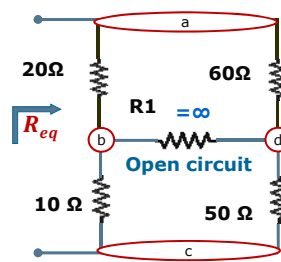


$$R_{eq} = (20 || 60) + (10 || 50)$$

$$= \frac{20 \cdot 60}{20 + 60} + \frac{10 \cdot 50}{10 + 50}$$

$$= 15 + 8.33$$

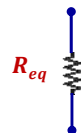
$$R_{eq} = 23.33 \Omega$$



$$R_{eq} = (20 + 10) || (60 + 50)$$

$$= \frac{30 \cdot 110}{30 + 110}$$

$$R_{eq} = 23.56 \Omega$$



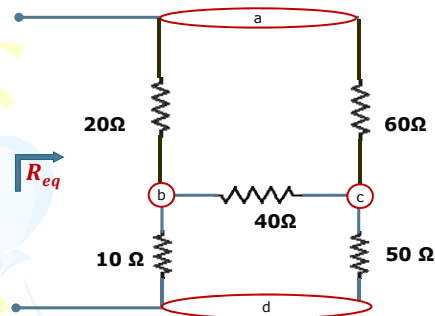
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Calculate R_{eq}

Identify if any resistance in series or parallel?

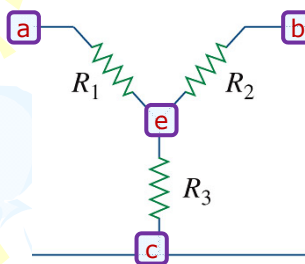


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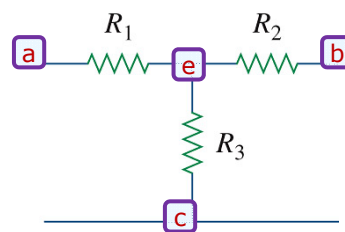
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Lecture #4 Wye or star or Tee network (Y - * - T)



(a)

WYE (Y) or Star



(b)

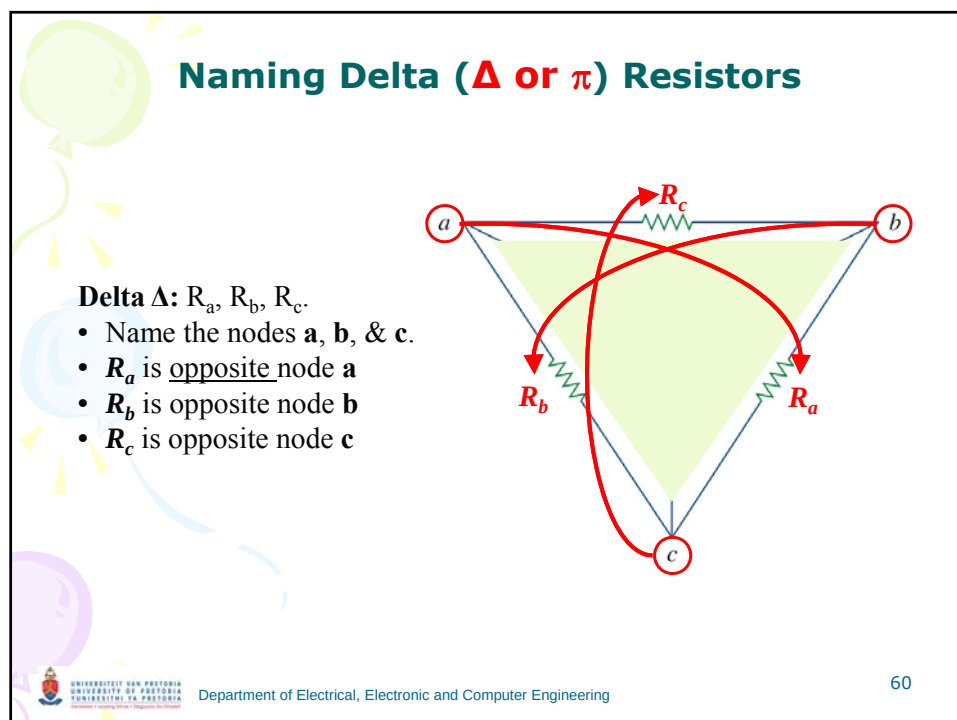
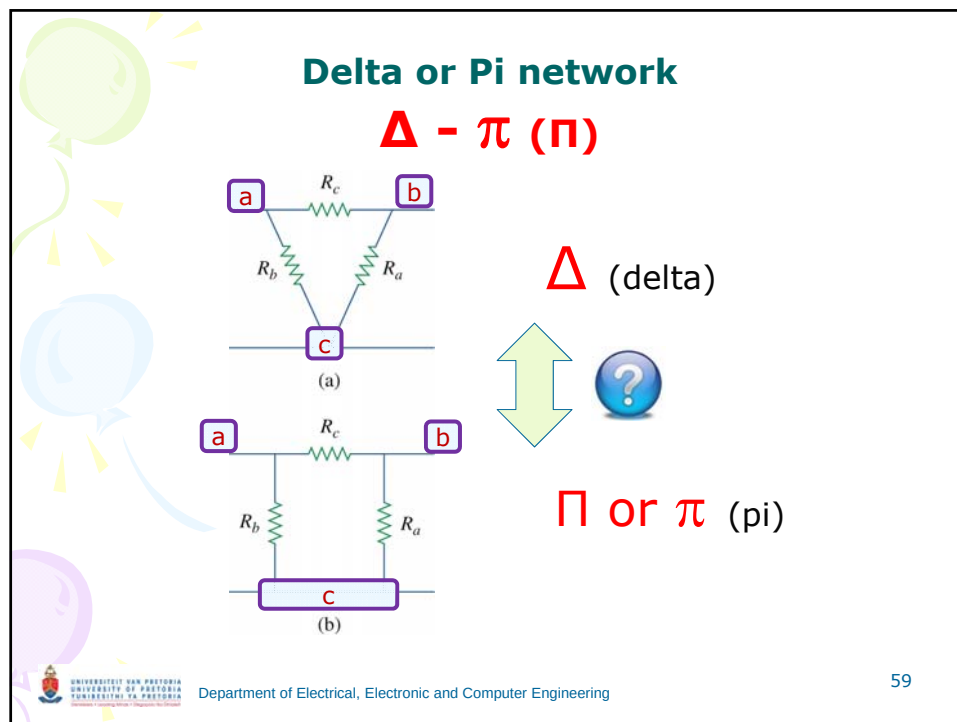
T Network



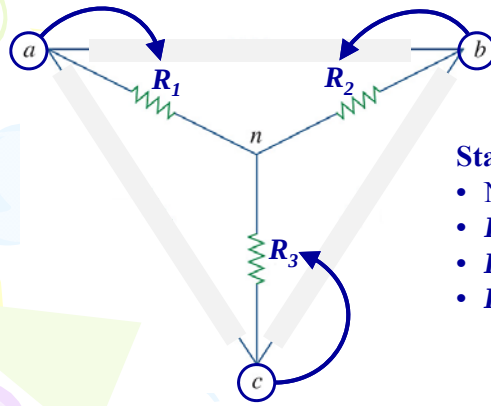
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Naming **Y**; T or star resistors



Star / Y: R_1, R_2, R_3 .

- Name the nodes **a**, **b**, and **c**.
- R_1 is connected to node **a**
- R_2 is connected to node **b**
- R_3 is connected to node **c**



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2.7 Wye-Delta Transformations

Y inside **Δ**

Delta → Wye

Δ → **Y**

Product of adjacent **Δ** R's

$$R_1 = \frac{R_b R_c}{(R_a + R_b + R_c)}$$

$$R_2 = \frac{R_c R_a}{(R_a + R_b + R_c)}$$

$$R_3 = \frac{R_a R_b}{(R_a + R_b + R_c)}$$

Wye → Delta

Y → **Δ**

All possible products of **Y** R's 2@t

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$



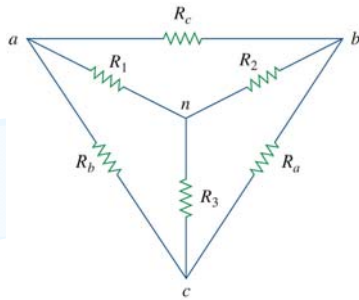
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2.7 Wye-Delta Transformations

Special Case: Balanced Y and Δ



$$R_a = R_b = R_c = R_{\Delta}$$

$$\text{then: } R_Y = \frac{R_{\Delta}}{3}$$

and

$$R_1 = R_2 = R_3 = R_Y$$

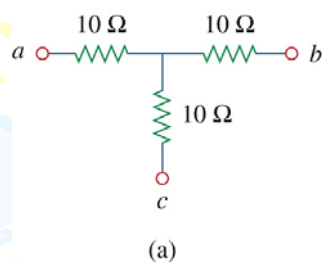
$$\text{then: } R_{\Delta} = 3R_Y$$



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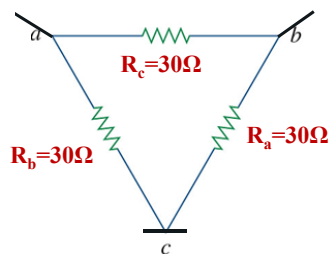
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P 2-48: Convert the circuits from Y to Δ :



$$R_1 = R_2 = R_3 = R_Y = 10\Omega$$

$$R_{\Delta} = 3R_Y = 3 \times 10 = 30\Omega$$

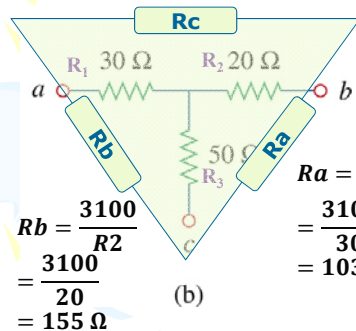


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P 2-48: Convert the circuits from Y to Δ :

$$R_c = \frac{3100}{R_3} = \frac{3100}{50} = 62 \Omega$$

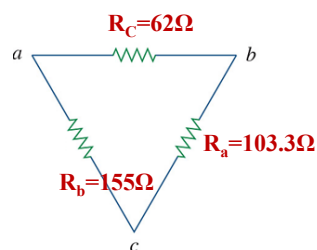


$$R_b = \frac{3100}{R_2} = \frac{3100}{20} = 155 \Omega$$

$$R_a = \frac{3100}{R_1} = \frac{3100}{30} = 103.3 \Omega$$

$$R_{\Delta} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_Y} = \frac{30 * 20 + 30 * 50 + 20 * 50}{R_Y}$$

$$R_{\Delta} = \frac{3100}{R_Y}$$

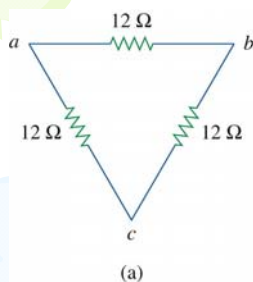


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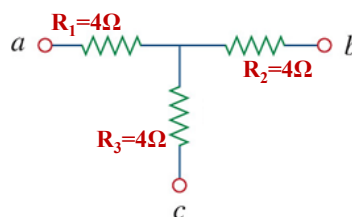
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P 2-49: Convert the circuits from Δ to Y :



$$R_a = R_b = R_c = R_{\Delta} = 12 \Omega$$

$$R_Y = \frac{1}{3} R_{\Delta} = \frac{12}{3} = 4 \Omega$$



Balanced delta

$$R_a = R_b = R_c = R_{\Delta}$$

$$R_Y = \frac{R_{\Delta}}{3}$$

Balanced Y

$$R_1 = R_2 = R_3 = R_Y$$

$$R_{\Delta} = 3 R_Y$$



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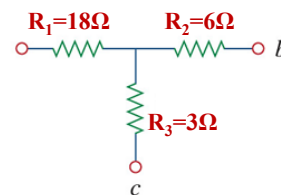
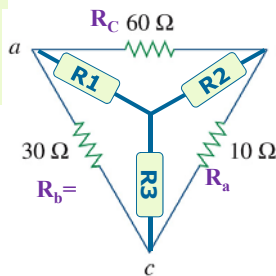
P 2-49: Convert the circuits from Δ to Y :

$$R_Y = \frac{R_x R_y}{R_a + R_b + R_c}; \quad R_a + R_b + R_c = 30 + 60 + 10 = 100$$

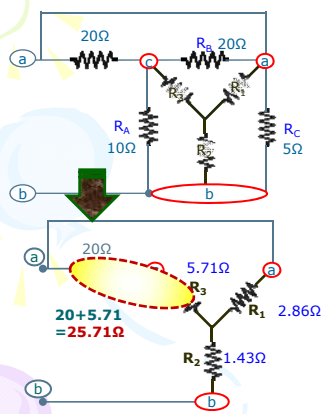
$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{30 * 60}{100} = 18 \Omega$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c} = \frac{10 * 60}{100} = 6 \Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{10 * 30}{100} = 3 \Omega$$

**P 2-44:** For the circuit in Figure, obtain the equivalent resistance at terminals a-b:**Method#1**

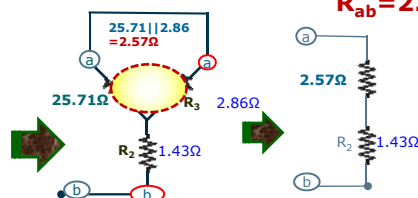
$\Delta \rightarrow Y$	$Y \rightarrow \Delta$
$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$	$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$
$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$	$R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$
$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$	$R_c = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$



$$R_1 = \frac{20 * 5}{20 + 10 + 5} = 2.86 \Omega$$

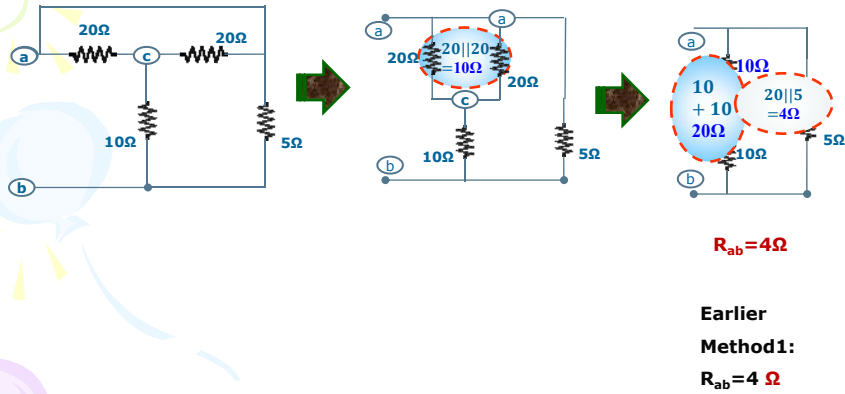
$$R_3 = \frac{20 * 10}{35} = 5.71 \Omega$$

$$R_2 = \frac{10 * 5}{35} = 1.43 \Omega$$



$$R_{ab} = 2.57 + 1.43 = 4 \Omega$$

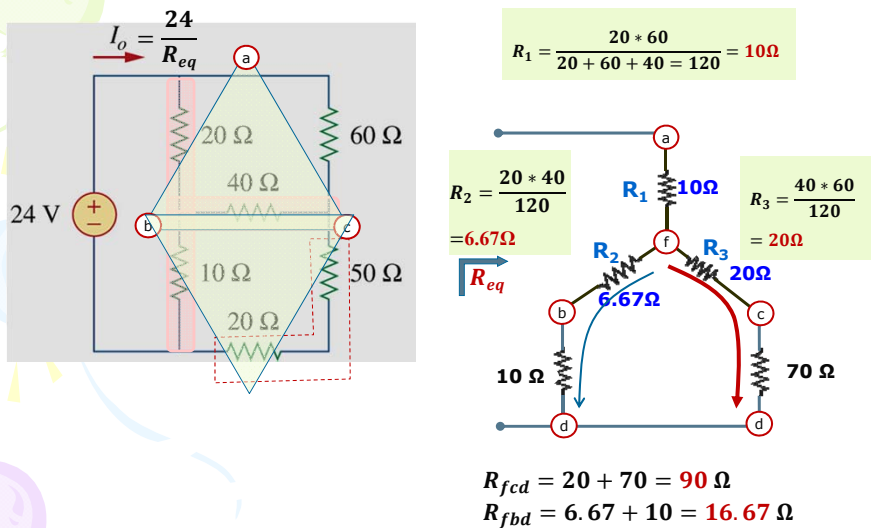
P 2-44: Method2: Calculate Resistance at terminals a-b



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EP 2.55: Calculate I_o in the circuit

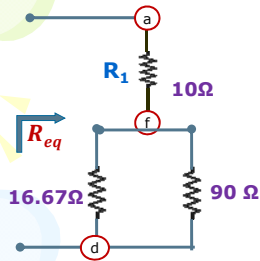


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Calculate I_o in the circuit

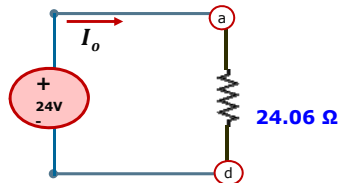


$$R_{fd} = 16.67 \parallel 90$$

$$= 14.06 \Omega$$

$$R_{ad} = 10 + 14.06$$

$$= 24.06 \Omega = R_{eq}$$



$$I_o = \frac{24}{24.06}$$

$$= 0.998 \text{ A}$$

Or 998 mA



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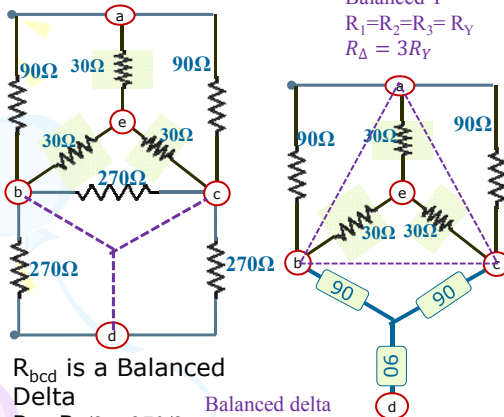
Calculate R_{eq} between node "ad"

Identify if any resistance in series or parallel?

None

R_{aebc} is a balanced Y.
 $R_{\Delta} = 3 \times R_Y = 3 \times 30 = 90 \Omega$

Balanced Y
 $R_1 = R_2 = R_3 = R_Y$
 $R_{\Delta} = 3R_Y$



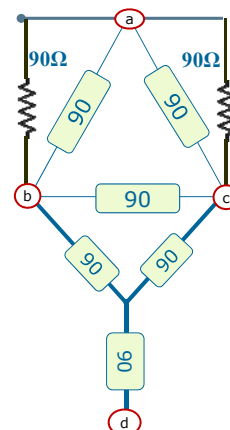
R_{bcd} is a Balanced Delta

$$R_Y = R_{\Delta} / 3 = 270 / 3$$

$$R_Y = 90 \Omega$$

Balanced delta
 $R_a = R_b = R_c = R_{\Delta}$
 $R_Y = \frac{R_{\Delta}}{3}$

$\Delta \rightarrow Y$	$Y \rightarrow \Delta$
$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$	$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$
$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$	$R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$
$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$	$R_c = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$



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Calculate R_{ad}

$90\Omega || 90\Omega = 45\Omega$

$\Delta_{bcf} \rightarrow Y_{bcgf}$
 $R_Y = \frac{1}{3} * 90$
 $= 30\Omega$

$\Delta \rightarrow Y$
$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$
$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$
$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$

$R_{ad} = (45 + 30) || (45 + 30) + (30 + 90)$
 $= (75 || 75) + 120$
 $= 37.5 + 120 = 157.5 \Omega$

157.5 Ω

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Calculate Req. between node "bc".

R_{aebc} is a balanced Y.
 $R_{\Delta} = 3 * R_Y = 3 * 30 = 90\Omega$

$\Delta \rightarrow Y$
$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$
$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$
$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$

$Y \rightarrow \Delta$
$R_a = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_3}$
$R_b = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2}$
$R_c = \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1}$

Identify, if any, resistance in series or parallel?
 $270 + 270 = 540\Omega$
 $540 || 270 = 180\Omega$

$90 || 180 = 60\Omega$
 $90 || 90 = 45\Omega$

$R_{bc} = (45 + 45) || 60$
 $= 90 || 60$
 $= 36\Omega$

36 Ω

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